

2

2	1
2026-02-25	1
1	1
2	2
3	2
4	2
	2
	2
	2
k	2
	2
	2
	2
Ramsey	2
5	2
6	2
	3
7	3
2026-03-06	3
1	3
	3
	3
2	3
	3
	4
3	4
4	4
5	4
6	4
7	5

XXXXXXXXXX 2026-02-25

1XXXXXXXXXXXXXXX

$\boxtimes G \boxtimes \boxtimes \boxtimes \boxtimes \boxtimes \boxtimes \boxtimes \boxtimes$

1. V
2. $E \otimes V$
3. V

$$V = \{v_1, v_2, v_3, \dots, v_n\}$$

$$E = \{(v_1, v_2), \dots, (v_4, v_5)\}$$

4. $\boxtimes \boxtimes \boxtimes \boxtimes G = (V, E)$
 1. $\boxtimes = |V| = n$
 2. $|E| = 0 \boxtimes \boxtimes$
 3. $|V| = 1 \boxtimes \boxtimes \boxtimes$
 4. 

5. 
5.  1.  1  2  0
2. 
3. 

4. $\{v_1, v_2\} \in E$ 5. $\{v_1, v_2\} \in E$ 6. $\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$ 7. $\{v_1, v_2\} \in E$

6. $\{v_1, v_2\} \in E$

1. $\{v_1, v_2\} \in E$

2. $\{v_1, v_2\} \in E$

3. $\{v_1, v_2\} \in E$

4. $\{v_1, v_2\} \in E$

5. $\{v_1, v_2\} \in E$

7. Δ, δ 8. Δ, δ

2. $\{v_1, v_2\} \in E$

1. $\{v_1, v_2\} \in E$ 2. $\{v_1, v_2\} \in E$

2. $\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$

eg. $\{v_1, v_2\} \in E$

3. $\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$ 1. $\{v_1, v_2\} \in E$

3. $\{v_1, v_2\} \in E$

$w(e) \in E$

4. $\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$

$|V_{K_n}| = n(n-1)/2$

$\{v_1, v_2\} \in E$

$m = n(n-1), \Delta^+ = \delta^+ = \delta^- = \Delta^- = n-1, \Delta = \delta = 2(n-1)$

$k - \{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$ $k \{v_1, v_2\} \in E$ $m = kn/2$

$\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$ 3 $\{v_1, v_2\} \in E$ 2- $\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$ $K_{m,n}$ $\{v_1, v_2\} \in E$

Ramsey $\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$

5. $\{v_1, v_2\} \in E$

1. $\{v_1, v_2\} \in E$

2. $\{v_1, v_2\} \in E$

3. $\{v_1, v_2\} \in E$

4. $\{v_1, v_2\} \in E$ $G[V'], G[E']$

6. $\{v_1, v_2\} \in E$

$\{v_1, v_2\} \in E$ $G_1 \cong G_2$

$\{v_1, v_2\} \in E$

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-
-
-
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eg. 4 3

eg. K_4  1 1 2 3 2 1 1 

XXXXXXXXXXXXXXXXXXXX

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- 
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7 XXXXX

1. $\boxtimes G_1 \cap G_2 = (V_1 \cap V_2, E_1 \cap E_2)$
2. $\boxtimes G_1 \cup G_2 = (V_1 \cup V_2, E_1 \cup E_2)$
3. $\boxtimes G_1 - G_2 = (V_1, E_1 - E_2)$
4. $\boxtimes \overline{G} = (V, E(K_n) - E)$
5. $\boxtimes \boxtimes G_1 \times G_2 = (V_1 \times V_2, E) \boxtimes E \boxtimes \boxtimes \boxtimes (u_1, v_1), (u_2, v_2) \in E \boxtimes \boxtimes u_1 = u_2 \boxtimes (v_1, v_2) \in E_2 \boxtimes v_1 = v_2 \boxtimes (u_1, u_2) \in E_1 \boxtimes$
6. $\boxtimes \boxtimes G \boxtimes \overline{G} \boxtimes$
7. $\boxtimes \boxtimes G_1 \oplus G_2 = (V_1 \cup V_2, E_1 \oplus E_2)$
8. $G - v \boxtimes \boxtimes \boxtimes v \boxtimes \boxtimes \boxtimes \boxtimes$
9. $G - e \boxtimes \boxtimes e$
10. $G + e \boxtimes \boxtimes e \boxtimes \boxtimes e \boxtimes \boxtimes \boxtimes \boxtimes$
11. $G + v \boxtimes \boxtimes \boxtimes v \boxtimes \boxtimes v \boxtimes \boxtimes \boxtimes \boxtimes$

XXXXXXXXXX 2026-03-06

11111111



$$G' = (V, E'), E' = \{(v, w) | (w, v) \in E\}$$



[illegible]

2 XXXXX

$$\begin{array}{ccccccc} \boxed{}\boxed{}\boxed{}\boxed{} & M & \boxed{}\boxed{}\boxed{}\boxed{} & n & \boxed{} & m & \boxed{}\boxed{}\boxed{}\boxed{}\boxed{}\boxed{}\boxed{}\boxed{} & n & \boxed{}\boxed{}\boxed{}\boxed{}\boxed{}\boxed{} & m & \boxed{}\boxed{}\boxed{}\boxed{} \end{array}$$

$$e_i v_i M_{ii} = 1 \quad 0 \leq i \leq n$$



1. $\sum_{i=1}^n b_{ij} = 2 \quad j = 1, 2, \dots, m$
2. $\sum_{j=1}^m b_{ij} = d(v_i) \quad i = 1, 2, \dots, n$
3. $\sum_{i,j} b_{ij} = 2m$
4. $2e_j \leq e_j$
5. $e_j \otimes e_k \leq e_j \otimes e_k$

6. $0 \leq v_i$

Example

$e_j \cdot v_i \cdot M_{ij} = 1 \cdot e_j \cdot v_i \cdot M_{ij} = -1 \cdot 0$

$D = \langle V, E \rangle \quad B(D) = (b_{ij})_{n \times m}$

$$b_{ij} = \begin{cases} 1, & v_i \leq e_j \\ 0, & v_i = e_j \\ -1, & v_i > e_j \end{cases}$$

Example

Example

3.3

$2 \leq m \quad A, B \quad e_k = (v_i, v_j) \quad A(k) = i, B(k) = j$

$G = (V, E) \quad V = \{v_1, v_2, v_3\} \quad E = \{(v_1, v_2), (v_2, v_3)\} \quad A = [1, 2] \quad B = [2, 3]$

Example

- $\mathcal{O}(nm) \Rightarrow \mathcal{O}(m)$
- $\mathcal{O}(nm) \Rightarrow \mathcal{O}(m)$

4.4

$0 \leq n-1 \quad B \quad A \quad n+1 \quad m$

(Forward Star)

v	head[v]
1	1
2	3
3	4
4	6
5	7 (null)

i	w	
1	v_2	5
2	v_3	2
3	v_4	8
4	v_1	1
5	v_5	3
6	—	—

Example

- 1 edges[1] edges[3-1]
- 2 edges[3] edges[4-1]
- 3 edges[4] edges[6-1]

Example

1. $d^+(v_i) = A(i+1) - A(i)$
2. $A(i) = \sum_{j=0}^{i-1} d^+(v_j)$
3. $\mathcal{O}(n+m)$
4. $\mathcal{O}(n+m)$
5. $\mathcal{O}(n+m)$

5.5

B

B

6.6

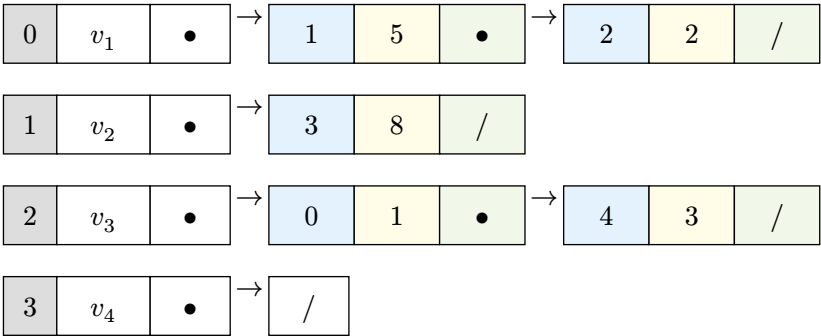
Example

- dst
- w

• link

Adjacency List

dst | w | link



data	firstedge
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dst	w	link
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$O(d^+(v)) \times O(d^-(v))$

7

- src
- dst
- weight
- d_link
- s_link